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Paper Code : PC-ECS301/PCC- CS301/PCC- CS85301/PCC- CS301/PCCS301 Data Structure & Algorithms

UPID : 003443

CS/B.TECH(N)/ODD/SEM-3/3443/2024-2025/1207

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What is the disadvantage of array data structure?
 - a) The amount of memory to be allocated should be known beforehand.
 - b) Elements of an array can be accessed in constant time.
 - c) Elements are stored in contiguous memory blocks.
 - d) Multiple other data structures can be implemented using arrays.
- (ii) Which of the following data structures allow Insertion and deletion from both ends?
 - a) Stack b) Deque c) Queue d) Strings
- (iii) Which of the following sorting algorithms provide the best time complexity in the worst-case scenario?
 - a) Merge Sort
 - b) Quick Sort
 - c) Bubble Sort
 - d) Selection Sort
- (iv) Worst case time complexity to access an element in a BST can be?
 - a) $O(n)$ b) $O(n * \log n)$ c) $O(1)$ d) $O(\log n)$
- (v) How are String represented in memory in C?
 - a) An array of characters.
 - b) The object of some class.
 - c) Same as other primitive data types.
 - d) LinkedList of characters.
- (vi) Which of the following data structures can be used to implement queues?
 - a) Stack b) Arrays c) Linked List d) Both b and c
- (vii) Which of the following is a Divide and Conquer algorithm?
 - a) Bubble Sort b) Selection Sort c) Heap Sort d) Merge Sort
- (viii) What is the best case time complexity of the binary search algorithm?
 - a) $O(1)$ b) $O(n)$ c) $O(\log 2n)$ d) $O(n^2)$
- (ix) What is the time complexity to insert an element to the front of a LinkedList (head pointer given)?
 - a) $O(n)$ b) $O(1)$ c) $O(\log n)$ d) $O(n * \log n)$
- (x) In a B+ tree, both the internal nodes and the leaves have keys.
 - a) True b) False
- (xi) What is the time complexity of an infix to postfix conversion algorithm?
 - a) $O(N \log N)$ b) $O(N)$ c) $O(N^2)$ d) $O(M \log N)$
- (xii) What is a hash table?
 - a) A structure that maps values to keys.
 - b) A structure that maps keys to values.
 - c) A structure used for storage.
 - d) A structure used to implement stack and queue.

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Write an algorithm to create linear linked list with n nodes. [5]
3. Write an algorithm to insert an element in Circular Queue. [5]
4. Justify the statement with proper example: All Binary Search Trees are Binary Tree but All Binary Trees are not Binary Search Tree. [5]
5. Find the time complexity of recursive algorithm of Tower of Hanoi. [5]

6. Convert the given Infix expression to postfix expression
 $A+B \cdot C-D \div F/H$

[5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. Create a AVL tree by inserting the following numbers in the order in which they are given:
17, 25, 19, 23, 75, 6. Explain it step wise with suitable diagram.
What is Threaded Binary Tree? What are its advantages? [8+3+4]
8. Define queue and its applications. Write an algorithm for insertion and deletion from queue. [5+5+5]
9. a) Define Circular Linked List and Double Linked List. [3+4+4+4]
b) Write an algorithm to delete an element from Double Linked List.
[Consider three cases: 1st node deletion, last node deletion and any intermediate node deletion]
10. a) Write a quick sort algorithm to sort a list. Explain with an example. [9+6]
b) Prove that the best case time complexity for quick sort is $O(n \log n)$ for Input size of n .
11. Write short notes on any three from following: [15]
a) AVL Tree
b) Dequeue and Priority Queue
c) Collision Resolution Techniques in Hashing
d) BFS

*** END OF PAPER ***

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